

Lab Report 6 - Cybersecurity Analytics

**Applying a Command Line Network Analysis Tool- tcpdump in
Analyzing Network Traffic**

Introduction

This lab is designed to work with Kali VM machine to get practical skills in network traffic capture, filtering, and analysis using tcpdump. tcpdump supports all major network-layer and transport-layer protocols, including how devices interact through protocols like TCP and UDP. TCP ensures reliable, connection-oriented communication, and UDP provides faster but connectionless data delivery (Easttom, 2022). Packet analysis provides important evidence for digital forensics. Packet analysis describes the process of capturing and interpreting live data as it flows across a network in order to better understand what is happening on that network (Sanders, 2017). In this lab, the Kali Linux OS is used to capture live packets using tcpdump on the eth0 interface while browsing two websites including MSUM website, and UND website.

Capture traffic from these websites for about 10-15 seconds and save it as *.pcap* file to observe TCP and UDP packets. This lab demonstrates appropriate use of packet capture filters, examine packet headers such as source and destination IP addresses, ports, flags, and sequence numbers to get better understanding of protocol level packet analysis which is essential in cybersecurity field.

PART A - TCP

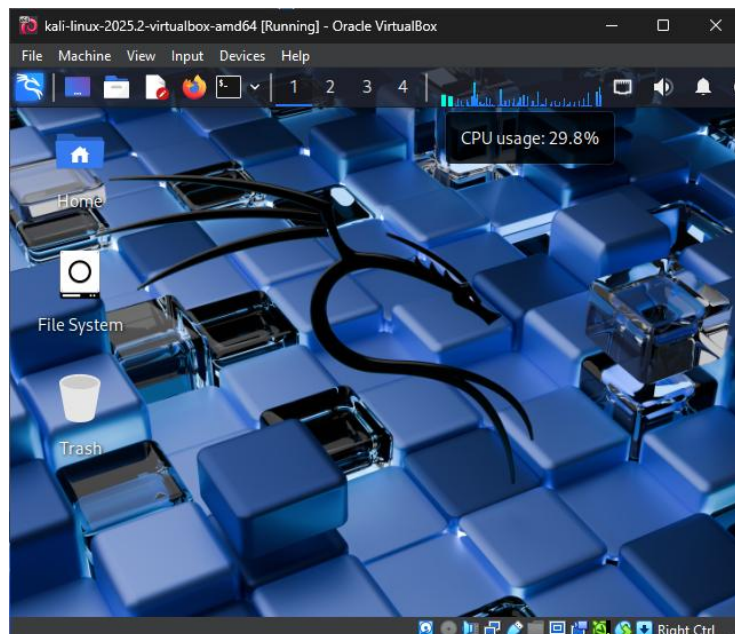
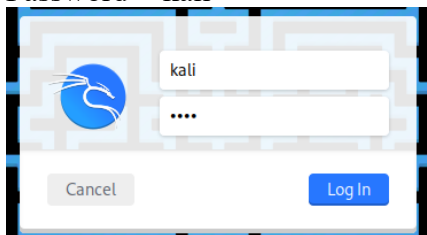
Step 1 - TCP Packet Capture

1. Logged to Azure Lab environment using following web link and logged into the Kali VM using Oracle VirtualBox.

https://labs.azure.com/virtualmachines?feature_vnext=true

Username - kali

Password - kali



2. I selected <https://www.mnstate.edu/> Moorhead University website to continue my lab work.
3. Used “whois” command to determine the IP address of the website in the terminal window in Kali. It shows the details of domain ownership.

```
File Actions Edit View Help
(kali@kali)-[~]
$ whois mnstate.edu
This Registry database contains ONLY .EDU domains.
The data in the EDUCAUSE Whois database is provided
by EDUCAUSE for information purposes in order to
assist in the process of obtaining information about
or related to .edu domain registration records.

The EDUCAUSE Whois database is authoritative for the
.EDU domain.

A Web interface for the .EDU EDUCAUSE Whois Server is
available at: http://whois.educause.edu

By submitting a Whois query, you agree that this information
will not be used to allow, enable, or otherwise support
the transmission of unsolicited commercial advertising or
solicitations via e-mail. The use of electronic processes to
harvest information from this server is generally prohibited
except as reasonably necessary to register or modify .edu
domain names.

Domain Name: MNSTATE.EDU

Registrant:
  Minnesota State University Moorhead
  Information Technology
  1104 7th Ave. So.
  Moorhead, MN 56563
  USA

Administrative Contact:
  Bryan Kotta
  Minnesota State University Moorhead
  1104 7th Ave So.
  Moorhead, MN 56563
  USA
  +1.2184772603
  domain@mnstate.edu

Technical Contact:
  Sam Groff
  Minnesota State University Moorhead
  1104 7th Ave So.
  Moorhead, MN 56563
  USA
  +1.2184772603
  domain@mnstate.edu

Name Servers:
  NS1-01.AZURE-DNS.COM
  NS4-01.AZURE-DNS.INFO
  NS3-01.AZURE-DNS.ORG
  MHDSEC.MNSTATE.EDU
  NS2-01.AZURE-DNS.NET
  MHD1.MNSTATE.EDU

Domain record activated: 16-Feb-1999
Domain record last updated: 03-Jun-2024
Domain expires: 31-Jul-2027
```

Then I used `nslookup mnstate.edu` command to get the IP address of targeted website and obtained 199.17.127.37 as IP address for mnstste.edu website.

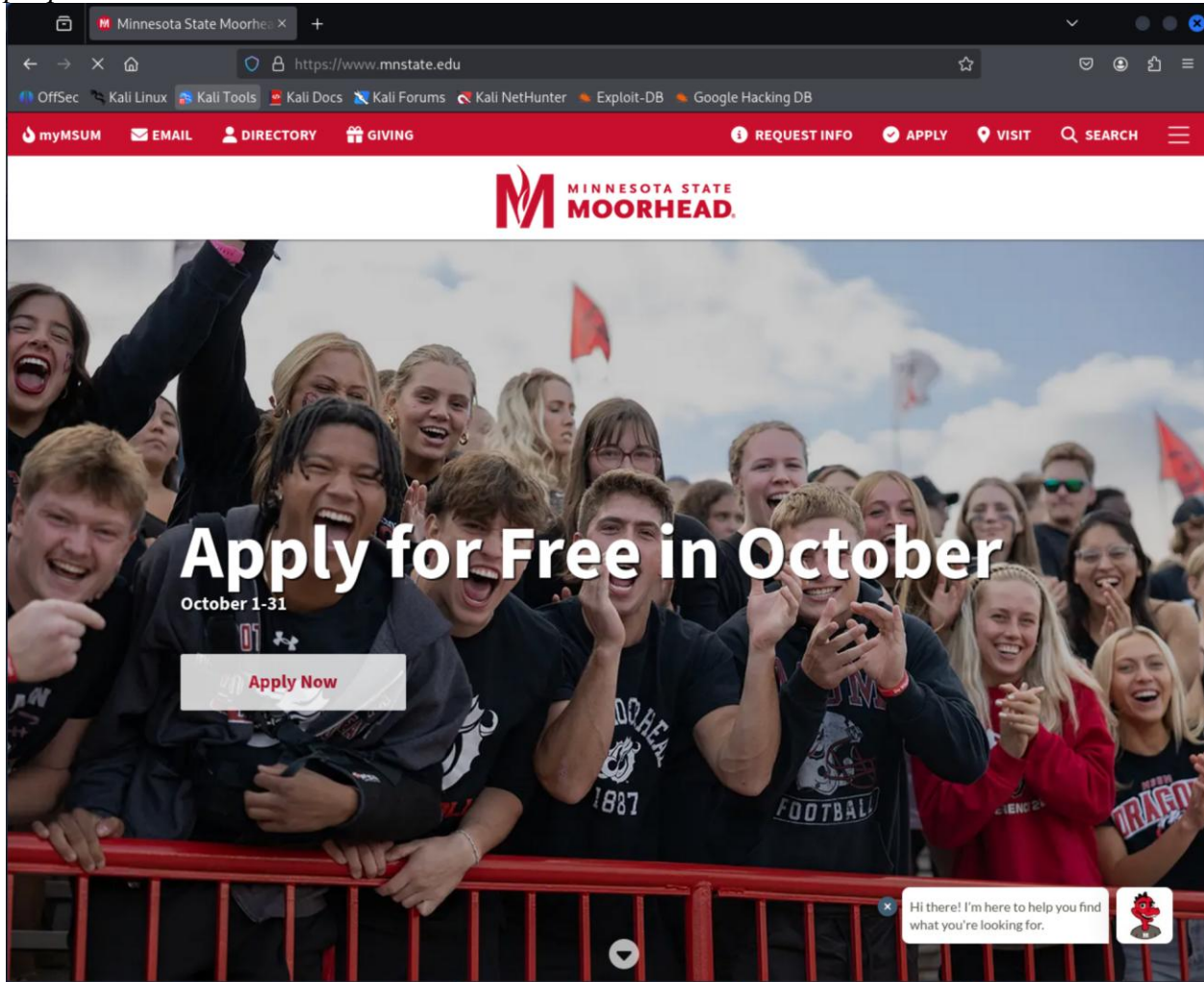
```
(kali@kali)-[~]
$ nslookup mnstate.edu
Server:      192.168.0.1
Address:     192.168.0.1#53

Non-authoritative answer:
Name:   mnstate.edu
Address: 199.17.127.37
```

4. Used `sudo tcpdump tcp and host 199.17.127.37 -w tcp_HashanK.pcap` command capture start to get packets from the Moorhead University website and write as `tcp_HashanK.pcap` file to launch tcpdump with a TCP filter.

```
(kali@kali)-[~]  
$ sudo tcpdump tcp and host 199.17.127.37 -w tcp_HashanK.pcap  
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

5. Opened the website in Firefox and waited few seconds to capture packets to dedicated `pcap` file.



Pressed `Ctrl + C` to stop the capture and observed 1247 packets captured during the session.

```
(kali@kali)-[~]  
$ sudo tcpdump tcp and host 199.17.127.37 -w tcp_HashanK.pcap  
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes  
^C1247 packets captured  
1247 packets received by filter  
0 packets dropped by kernel
```


Step 2 - Analyzing the capture - TCP

1. Located the capture file using *ls* command and found that it is stored in current working Directory.

```
(kali@kali)-[~]
$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  tcp_HashanK.pcap  Templates  Videos
```

2. Used *sudo tcpdump -r tcp_HashanK.pcap -c 10* command to display first 10 TCP packets from *tcp_HashanK.pcap* file.

```
(kali@kali)-[~]
$ sudo tcpdump -r tcp_HashanK.pcap -c 10
[sudo] password for kali:
reading from file tcp_HashanK.pcap, link-type EN10MB (Ethernet), snapshot length 262144
17:54:01.303243 IP 10.0.2.15.33694 > 199.17.127.37.https: Flags [S], seq 1020413031, win 64240, options [mss 1460,sackOK,TS val 419517848 ecr 0,nop,wscale 7], length 0
17:54:01.303934 IP 10.0.2.15.33696 > 199.17.127.37.https: Flags [S], seq 466090329, win 64240, options [mss 1460,sackOK,TS val 419517849 ecr 0,nop,wscale 7], length 0
17:54:01.341144 IP 199.17.127.37.https > 10.0.2.15.33694: Flags [S.], seq 48128001, ack 1020413032, win 65535, options [mss 1460], length 0
17:54:01.341250 IP 10.0.2.15.33694 > 199.17.127.37.https: Flags [.], ack 1, win 64240, length 0
17:54:01.341664 IP 199.17.127.37.https > 10.0.2.15.33696: Flags [S.], seq 48192001, ack 466090330, win 65535, options [mss 1460], length 0
17:54:01.341675 IP 10.0.2.15.33696 > 199.17.127.37.https: Flags [.], ack 1, win 64240, length 0
17:54:01.343500 IP 10.0.2.15.33696 > 199.17.127.37.https: Flags [P.], seq 1:664, ack 1, win 64240, length 663
17:54:01.343787 IP 199.17.127.37.https > 10.0.2.15.33694: Flags [.], ack 664, win 65535, length 0
17:54:01.344187 IP 10.0.2.15.33696 > 199.17.127.37.https: Flags [P.], seq 1:664, ack 1, win 64240, length 663
17:54:01.344430 IP 199.17.127.37.https > 10.0.2.15.33696: Flags [.], ack 664, win 65535, length 0
```

- Select first packet and get the details using *tcpdump -r tcp_HashanK.pcap -v -c 1* command.

```
(kali@kali)-[~]
$ tcpdump -r tcp_HashanK.pcap -v -c 1
reading from file tcp_HashanK.pcap, link-type EN10MB (Ethernet), snapshot length 262144
17:54:01.303243 IP (tos 0x0, ttl 64, id 53273, offset 0, flags [DF], proto TCP (6), length 60)
10.0.2.15.33694 > 199.17.127.37.https: Flags [S], cksum 0x5274 (incorrect -> 0x859d), seq 1020413031, win 64240, options [mss 1460,sackOK,TS val 419517848 ecr 0,nop,wscale 7], length 0

(kali@kali)-[~]
$ tcpdump -r tcp_HashanK.pcap -vv -c 1
reading from file tcp_HashanK.pcap, link-type EN10MB (Ethernet), snapshot length 262144
17:54:01.303243 IP (tos 0x0, ttl 64, id 53273, offset 0, flags [DF], proto TCP (6), length 60)
10.0.2.15.33694 > 199.17.127.37.https: Flags [S], cksum 0x5274 (incorrect -> 0x859d), seq 1020413031, win 64240, options [mss 1460,sackOK,TS val 419517848 ecr 0,nop,wscale 7], length 0

(kali@kali)-[~]
$ tcpdump -r tcp_HashanK.pcap -vvv -c 1
reading from file tcp_HashanK.pcap, link-type EN10MB (Ethernet), snapshot length 262144
17:54:01.303243 IP (tos 0x0, ttl 64, id 53273, offset 0, flags [DF], proto TCP (6), length 60)
10.0.2.15.33694 > 199.17.127.37.https: Flags [S], cksum 0x5274 (incorrect -> 0x859d), seq 1020413031, win 64240, options [mss 1460,sackOK,TS val 419517848 ecr 0,nop,wscale 7], length 0
```

- Checked the IP address of the VM using *ifconfig* command.

```
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
inet6 fd17:625c:f037:2:818c:c8e:2aca:4ffe prefixlen 64 scopeid 0x0<global>
inet6 fe80::2873:fc42:5e1:7c15 prefixlen 64 scopeid 0x20<link>
ether 08:00:27:d1:f8:5d txqueuelen 1000 (Ethernet)
RX packets 17619 bytes 19910389 (18.9 MiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 5745 bytes 755667 (737.9 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

3. Extract the following information from the first packet.
 - a. Timestamp - 17:54:01.303243
 - b. Source IP, Source Port - 10.0.2.15:33694
 - c. Destination IP, Destination port - 199.17.127.37:443 (https)
 - d. Flag information - Flag [S] - SYN Flag
 - e. Sequence number - 1020413031
 - f. Acknowledgement number - No ACK
 - g. Window size - 64240
 - h. Data Length - 0

PART B - UDP

Step 3 - UDP Packets Capture

1. I selected und.edu University of North Dakota website to continue my lab work.
2. Used “whois” command to determine the IP address of the website in the terminal window in Kali. It shows the details of domain ownership.

```
(kali㉿kali)-[~]
$ whois und.edu
This Registry database contains ONLY .EDU domains.
The data in the EDUCAUSE Whois database is provided
by EDUCAUSE for information purposes in order to
assist in the process of obtaining information about
or related to .edu domain registration records.

The EDUCAUSE Whois database is authoritative for the
.EDU domain.

A Web interface for the .EDU EDUCAUSE Whois Server is
available at: http://whois.educause.edu

By submitting a Whois query, you agree that this information
will not be used to allow, enable, or otherwise support
the transmission of unsolicited commercial advertising or
solicitations via e-mail. The use of electronic processes to
harvest information from this server is generally prohibited
except as reasonably necessary to register or modify .edu
domain names.

-----

Domain Name: UND.EDU

Registrant:
  University of North Dakota
  264 Centennial Drive, STOP 8176
  NDUS Information Technologies
  Grand Forks, ND 58202-8176
  USA

Administrative Contact:
  Madhavi Marasinghe
  University of North Dakota
  Chief Information Officer
  264 Centennial Drive, STOP 8176
  Grand Forks, ND 58202-8176
  USA
  +1.7017775756
  und.enterprise.services@email.und.edu

Technical Contact:
  UND-DNS
  University of North Dakota
  4349 James Ray Drive
  Grand Forks, ND 58202
  USA
  +1.7017773171
  ndus.dns.tech@ndus.edu

Name Servers:
  NS1.EXT.NDUS.EDU
  NS2.EXT.NDUS.EDU

Domain record activated: 08-Jun-1996
Domain record last updated: 02-Jul-2025
Domain expires: 31-Jul-2026
```

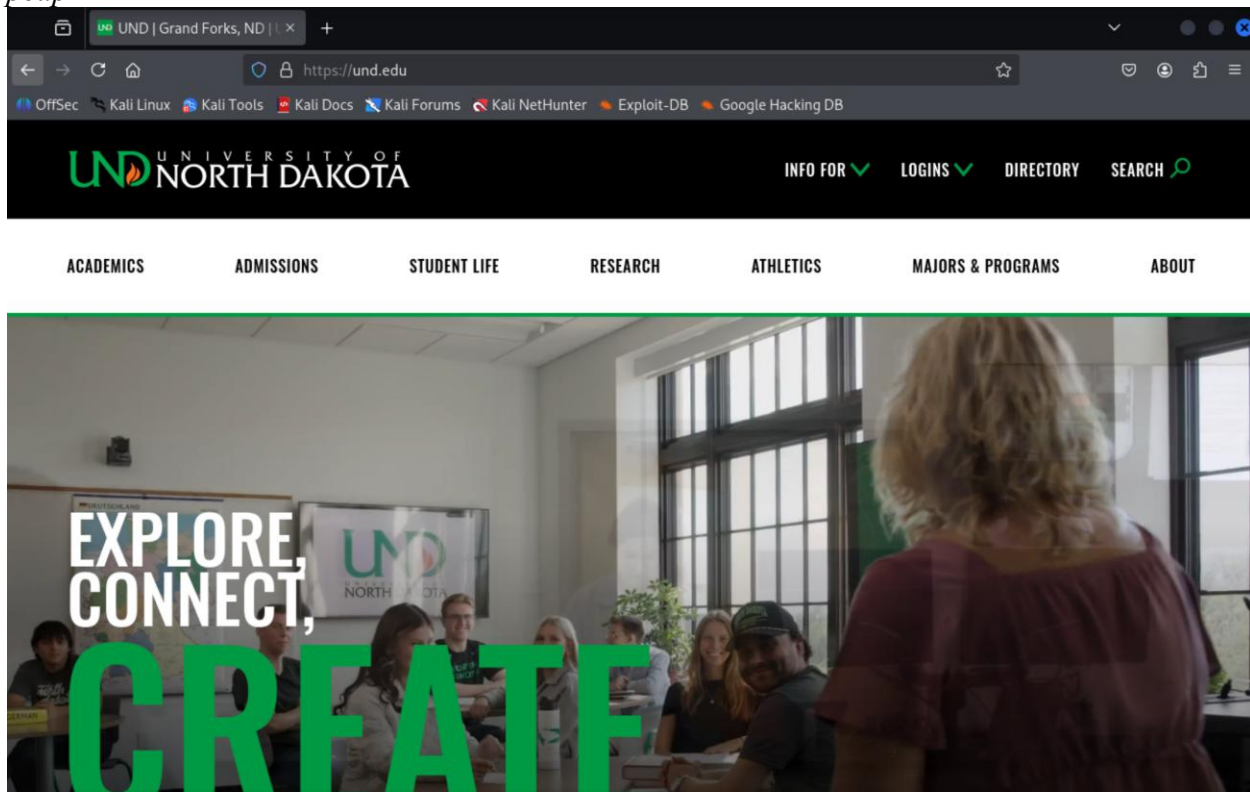
Then I used *nslookup und.com* command to get the IP address of targeted website and obtained 34.96.123.237 as IP address for University of North Dakota website.

```
(kali㉿kali)-[~]  
$ nslookup und.com  
Server:      192.168.0.1  
Address:     192.168.0.1#53  
  
Non-authoritative answer:  
Name:   und.com  
Address: 34.96.123.237
```

3. Used *sudo tcpdump udp -w udp_HashanK.pcap* command capture start to get packets from the Northland College website and write as *udp_HashanK.pcap* file to launch tcpdump with a TCP filter.

```
(kali㉿kali)-[~]  
$ sudo tcpdump udp -w udp_HashanK.pcap  
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes  
^
```

4. Opened the website in Firefox and waited few seconds to capture packets to dedicated *pcap* file.



Pressed *Ctrl + C* to stop the capture and observed 252 packets captured during the session.

```
(kali㉿kali)-[~]  
$ sudo tcpdump udp -w udp_HashanK.pcap  
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes  
^C252 packets captured  
252 packets received by filter  
0 packets dropped by kernel
```


Step 4 - Analyzing the capture - UDP

1. Located the capture file using *ls* command and found that it is stored in current working Directory.

```
(kali@kali)-[~]
$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  tcp_HashanK.pcap  Templates  udp_HashanK.pcap  Videos
```

2. Used *sudo tcpdump -r udp_HashanK.pcap -c 5* command to display first 5 UDP packets from *udp_HashanK.pcap* file.

```
(kali@kali)-[~]
$ sudo tcpdump udp -r udp_HashanK.pcap -c 5
reading from file udp_HashanK.pcap, link-type EN10MB (Ethernet), snapshot length 262144
20:03:17.497415 IP 10.0.2.15.41042 > puma7-atom.ht.home.domain: 37166+ A? und.edu. (25)
20:03:17.504858 IP puma7-atom.ht.home.domain > 10.0.2.15.41042: 37166 1/0/0 A 134.129.183.70 (41)
20:03:23.415285 IP 10.0.2.15.60888 > puma7-atom.ht.home.domain: 64966+ A? www.googletagmanager.com. (42)
20:03:23.415527 IP 10.0.2.15.48722 > puma7-atom.ht.home.domain: 37627+ A? googleads.g.doubleclick.net. (45)
20:03:23.416460 IP 10.0.2.15.57593 > puma7-atom.ht.home.domain: 2087+ A? connect.facebook.net. (38)
```

- Select first packet and get the details using *tcpdump -r tcp_HashanK.pcap -v -c 1* command.

```
(kali@kali)-[~]
$ sudo tcpdump udp -r udp_HashanK.pcap -v -c 1
reading from file udp_HashanK.pcap, link-type EN10MB (Ethernet), snapshot length 262144
20:03:17.497415 IP (tos 0x0, ttl 64, id 29495, offset 0, flags [DF], proto UDP (17), length 53)
    10.0.2.15.41042 > puma7-atom.ht.home.domain: 37166+ A? und.edu. (25)

(kali@kali)-[~]
$ sudo tcpdump udp -r udp_HashanK.pcap -vv -c 1
reading from file udp_HashanK.pcap, link-type EN10MB (Ethernet), snapshot length 262144
20:03:17.497415 IP (tos 0x0, ttl 64, id 29495, offset 0, flags [DF], proto UDP (17), length 53)
    10.0.2.15.41042 > puma7-atom.ht.home.domain: [bad udp cksum 0xccea -> 0x2489!] 37166+ A? und.edu. (25)

(kali@kali)-[~]
$ sudo tcpdump udp -r udp_HashanK.pcap -vvv -c 1
reading from file udp_HashanK.pcap, link-type EN10MB (Ethernet), snapshot length 262144
20:03:17.497415 IP (tos 0x0, ttl 64, id 29495, offset 0, flags [DF], proto UDP (17), length 53)
    10.0.2.15.41042 > puma7-atom.ht.home.domain: [bad udp cksum 0xccea -> 0x2489!] 37166+ A? und.edu. (25)
```

- Checked the IP address of the VM using *ifconfig* command.

```
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:818c:c8e:2aca:4ffe prefixlen 64 scopeid 0<global>
    inet6 fe80::2873:fc42:5e1:7c15 prefixlen 64 scopeid 0<20<link>
    ether 08:00:27:d1:f8:5d txqueuelen 1000 (Ethernet)
    RX packets 17619 bytes 19910389 (18.9 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 5745 bytes 755667 (737.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

3. Extract the following information from the first packet.
 - a. Timestamp - 20:03:17.497415
 - b. Layer 3 protocol listed - IP
 - c. Source IP, Source Port - 10.0.2.15:41042
 - d. Destination IP, Destination port - puma7-atom.ht.home.domain:53 (DNS)
 - e. Layer 4 protocol listed - UDP
 - f. Data Length - 25

PART C - TOOLS

Step 5 - Comparing Tools

Criteria	tcpdump	Wireshark
Operating Systems Support	Linux, UNIX, Windows, macOS	Linux, Windows, macOS
Interface type	Command Line Interface (CLI)	Graphical User Interface (GUI)
Protocols	Network layer and Transport Layer Protocols (Ethernet, ARP, IPv4/IPv6, TCP, UDP, ICMP, DNS, HTTP, HTTPS)	captures network frames at the physical and data link layers (Layers 1 and 2) Presents the captured data by showing the different layers, from Layer 2 up to Layer 7
Statistical Features	Basic statistics (Timestamps, Packet counts, Protocol types, Flags and etc.)	Advanced statistics and analysis tools: I/O graphs, conversations, endpoints, flow charts, expert diagnostics, and protocol hierarchies
Analytical Features	Basic analysis, ideal for quick summaries or simple filters	Advanced features, including complex filtering, stream reassembly, protocol graphs, and detailed packet dissection
Output Formats	Saves to .pcap files	Saves to .pcap and .pcapng file and able to read and write both formats
Limitations	Does not support for GUI	Cannot get better output with low resource environment

Conclusion

This lab helps to cover the practical aspects of the live TCP and UDP protocol packet analysis by using tcpdump. First, I visited two websites and captured live TCP and UDP packets and saved them as .pcap files. By analyzing packets, I got experience of using packet capture filters, examining packet headers such as source and destination IP addresses, ports, flags, and sequence numbers, data length of TCP and UDP protocol packets.

The TCP analysis highlighted the reliability features of the protocol, including its three-way handshake to establish a session, sequence and acknowledgment numbers, window size, and connection control flags, which ensure ordered and verified data delivery between hosts. The UDP capture primarily observed through DNS query packets. It shows the efficiency of connectionless communication, where data is transmitted without prior session establishment or acknowledgment.

References

1. Sanders, C. (2017). *Practical packet analysis* (3rd ed.).
2. Minnesota State University Moorhead. (n.d.). *Minnesota State University Moorhead*. Retrieved October 5, 2025, from <https://www.mnstate.edu/>
3. University of North Dakota. (n.d.). *University of North Dakota*. Retrieved October 5, 2025, from <https://und.edu/>
4. Easttom, C. (2022). *Computer Security Fundamentals* (5th ed.). Pearson Education.
5. Wireshark Foundation. (2024). *Wireshark User's Guide – Network Protocol Analyzer*. Wireshark Foundation. <https://www.wireshark.org/docs/>
6. tcpdump. (n.d.). *tcpdump*. Retrieved October 5, 2025, from <https://www.tcpdump.org/>